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# FLUID PHASE EQUILIBRIUM STUDIES ON BINARY AND TERNARY MIXTURES OF CARBON DIOXIDE WITH HEXADECANE, 1-DODECANOL, 1,8-OCTANEDIOL AND DOTRIACONTANE AT 393.2 K AND AT PRESSURES UP TO 100 MPa

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ABSTRACT

Spee M. and Schneider G. M., 1990. Fluid Phase Equilibrium Studies on Binary and Ternary Mixtures of Carbon Dioxide with Hexadecane, 1-Dodecanol, 1,8-Octanediol and Dotriacontane at 393.2 K and at Pressures up to 100 MPa

Isothermal p(w) diagrams were determined for the binary systems $CO_2$ + hexadecane, $CO_2$ + 1-dodecanol, $CO_2$ + 1,8-octanediol, $CO_2$ + dotriacontane and hexadecane + 1,8-octanediol at 393.2 K and pressures from 10 to 100 MPa. In addition isothermal measurements were performed on the ternary systems $CO_2$ + hexadecane + 1-dodecanol, $CO_2$ + hexadecane + 1,8-octanediol, $CO_2$ + 1-dodecanol + 1,8-octanediol and hexadecane + 1-dodecanol + 1,8-octanediol at the same conditions; an isobar of the system $CO_2$ + hexadecane +1-dodecanol at 25 MPa was also measured.

## INTRODUCTION

Multicomponent fluid phase equilibrium data and the understanding of the solubility of low-volatile components in supercritical solvents are of increasing interest for applications in Supercritical Fluid Extraction (SFE) where compressed supercritical gases are used for the extraction of low-volatile substances. Compressed carbon dioxide has a high solvent power for many low-volatile components and up to now it is the most important supercritical solvent for practical applications (see e.g. Schneider et al., 1980, 1987, 1988).

A high solubility is usually combined with a poor selectivity. Therefore the use of a moderator is necessary in many cases in order to increase solvent power and/or selectivity. The effects of gas mixtures (Katzenski-Ohling et al., 1986, 1987) and volatile entrainers (Hölscher et al., 1988, 1989, Brunner et al., 1983) have been described earlier.

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Binary and ternary systems of the type $CO_2$ + alkane and $CO_2$ + alcohol can be used for the comparison of the solubility behaviour of a polar and an unpolar low-volatile component in compressed carbon dioxide. Since the systems $CO_2$ + hexadecane and $CO_2$ + 1-dodecanol are very similar the ternary system $CO_2$ + hexadecane + 1-dodecanol is of high interest for a separation process. The other systems are part of two quaternary systems, every quaternary system being made up of four ternary and six binary systems.

## EXPERIMENTAL

Experimental data were obtained by using an analytical high-pressure cell. A detailed description of the experimental apparatus is given in earlier papers by Konrad et al. (1982, 1983) and Deiters et al. (1986).

The measuring technique is briefly described in the following. The low-volatile components were filled into the pressure vessel (volume of the vessel about 100 $cm^3$). Pressure was increased by addition of compressed carbon dioxide or hexadecane via a hand pump. The pressure was measured to within ±0.1 MPa by a Heise gauge and temperature within ±0.2 K by a thermocouple placed in the fluid mixture.

The $p,T,w',w''$ ($w$ = mass fraction) equilibrium data were determined by sampling about 50 mg from both phases. During sampling the sample splits into two phases in the low pressure branch of the apparatus. The mass of the gaseous component ($CO_2$) can be calculated from an equation of state (in this case the van der Waals EOS) with volume, pressure and temperature of the sampling system being known. The low-volatile components were dissolved from the outlet capillary tubes by a solvent and analysed by gas chromatography using an external standard. In the present measurements the mass fractions could be determined to $|\Delta w| < 0.015$ (for details of the error analysis see Spee (1990)).

The liquid components were purchased by Riedel de Haën AG (hexadecane >99.5%), Fluka AG (1-dodecanol >99.5%, tetradecane >99%) and Janssen GmbH (1,8-octanediol >99%, dotriacontane >97%). These substances were used without any further purification.

## RESULTS AND DISCUSSION

In the present paper data for five binary and four ternary isothermal pressure-composition diagrams as well as for one binary and one ternary isobaric temperature-composition diagram are presented. The measurements were performed between 10 MPa and 100 MPa, usually between 10 MPa and the upper critical pressure.

carbon dioxide + hexadecane

The isothermal p(w) phase diagram of the system carbon dioxide and hexadecane at 393.2 K is presented in figure 1a. The mutual miscibility increases with increasing pressure. The critical pressure at 393.2 K is $p_c$(393.2 K) = (25.6± 0.2) MPa, the critical mass fraction is $w_c(CO_2)$ = (0.70 ± 0.02). This system corresponds to type III of the classification of van Konynenburg and Scott with a continuous critical line from the liquid-liquid to the liquid-gas phase equilibrium. Below 25.6 MPa the heterogeneous region has got the shape of a distorted hyperbol.

At 25 MPa the isobaric T(w) section (fig. 2b) exhibits an upper and a lower critical temperature ($T^c_u$(25 MPa) = (298± 2) K, $T^c_l$ (25 MPa) = (384± 1) K). This leads to a very interesting ternary phase behaviour in the system carbon dioxide + hexadecane + 1-dodecanol (fig. 4).

carbon dioxide + 1-dodecanol

The projection of the critical line of the system carbon dioxide + 1-dodecanol on the pressure-temperature plane is very similar to that of the system carbon dioxide + hexadecane. Figure 1b shows the p(w) phase diagram of this system at 393.2 K. The upper critical pressure at 393.2 K is $p_c$(393.2 K) = (27.5±0.2) MPa and thus higher than in the system carbon dioxide + hexadecane. The critical mass fraction is $w_c(CO_2)$ = (0.68 ± 0.02).

The isobaric T(w) section at 25 MPa is shown in figure 2b. Over the whole temperature interval the two-phase region looks like a band.

carbon dioxide + 1,8-octanediol

The system carbon dioxide + 1,8-octanediol differs from the other carbon dioxide systems. No critical point exists at 393.2 K. The mutual miscibility is small over the whole pressure interval (fig. 1c). The shape of the heterogeneous region indicates a liquid-liquid phase equilibrium.

Octanediol is a strongly polar molecule with two hydroxyl-groups, the melting point is 333 K. The low mutual miscibility with $CO_2$ can be explained from a high degree of association in the liquid phase.

carbon dioxide + dotriacontane

Dotriacontane ($C_{32}H_{66}$) is an alkane of the molar mass 450.9 g $mol^{-1}$. The system carbon dioxide + dotriacontane resembles the systems carbon dioxide + hexadecane and carbon dioxide + 1-dodecanol. Only the upper critical pressure is shifted to higher values. Thus the mutual miscibility is reduced. The critical pressure is $p_c$(393.2 K) = (65±0.2) MPa with a critical mass fraction of $w_c(CO_2)$ = (0.74 ± 0.02) (fig. 1d).

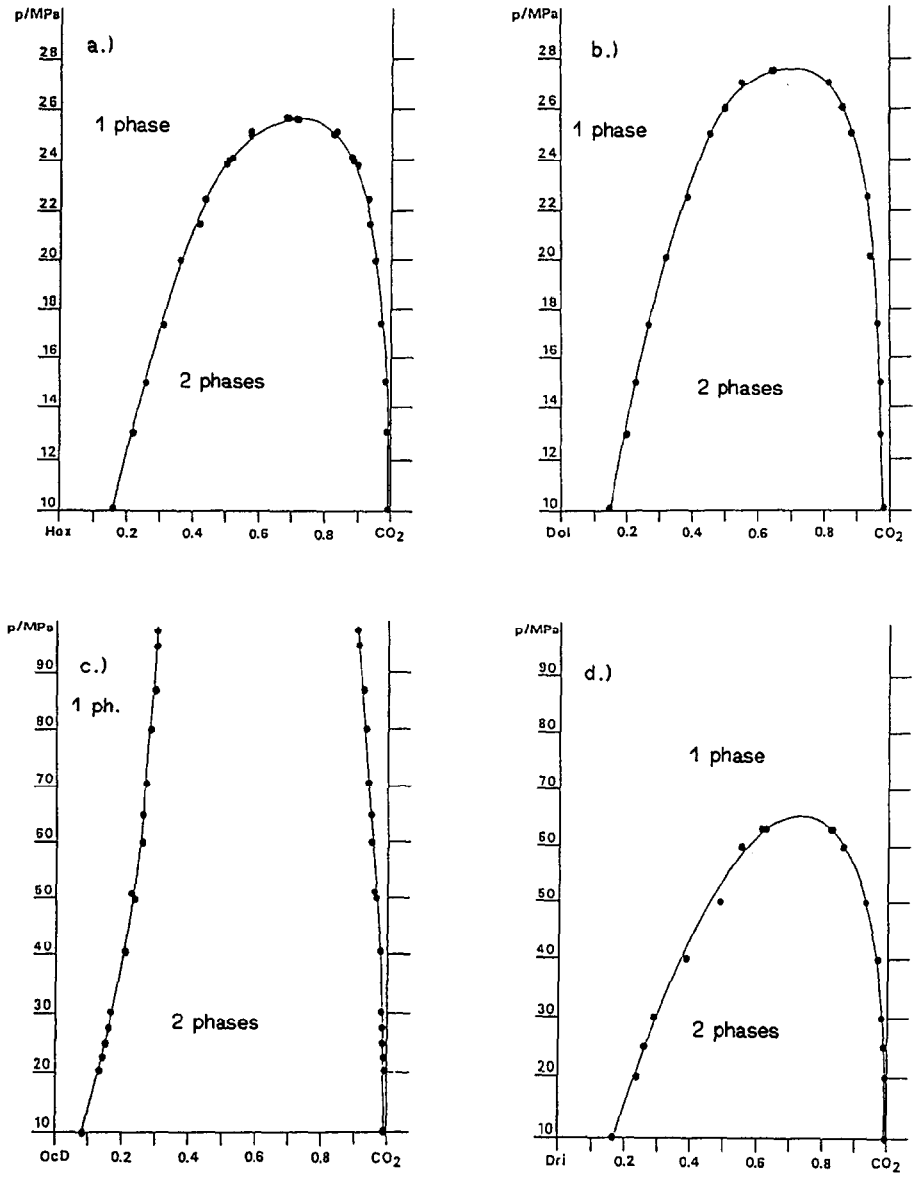



Fig. 1 p(w)-diagrams for binary $CO_2$ + alkane/alcohol systems:
(a) $CO_2$ + hexadecane at 393.2 K; (b) $CO_2$ + 1-dodecanol at 393.2 K
(c) $CO_2$ + 1,8-octanediol at 393.2 K; (d) $CO_2$ + dotriacontane at 393.2 K

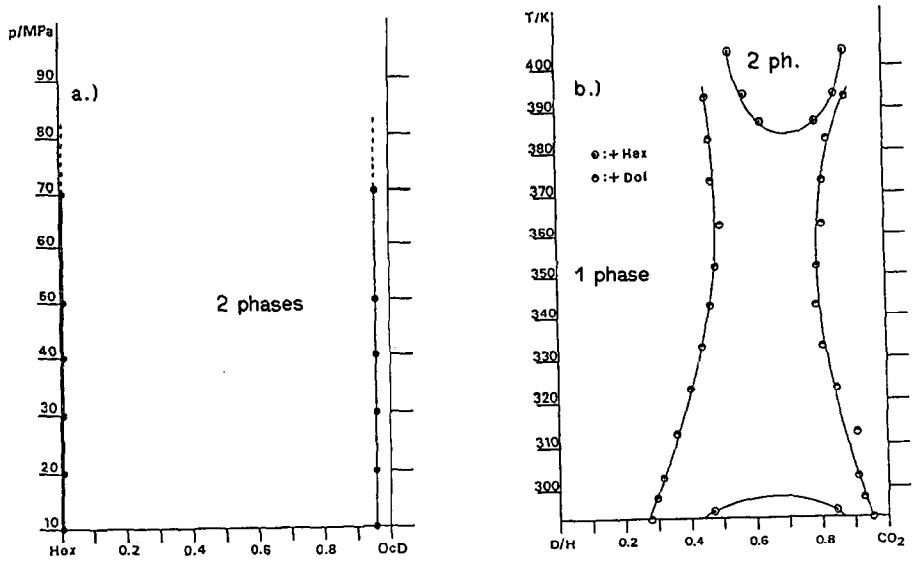



Fig. 2 p(w)- or T(w)-diagrams for binary $CO_2$ + alkane/alcohol systems;
(a) hexadecane + 1,8-octanediol at 393.2 K;
(b) $CO_2$ + hexadecane and carbon dioxide + 1-dodecanol at 25 MPa

hexadecane + 1,8-octanediol

Hexadecane is slightly better soluble in 1,8-octanediol (4 mass-%) than 1,8-octanediol in hexadecane (1 mass-%). This behaviour is independent of pressure. In figure 2a the heterogeneous area covers 95% of the diagram.

Pressure was increased in this system by a hand pump filled with hexadecane. Each capillary of the pressurizing system had to be heated up to 333 K to avoid reaching the melting curve.

carbon dioxide + hexadecane + 1-dodecanol

The ternary system carbon dioxide + hexadecane + 1-dodecanol is presented in an isothermal (at 393.2 K) (fig. 3) and an isobaric (at 25 MPa) (fig. 4) representation. A separation between hexadecane and 1-dodecanol in carbon dioxide did not occur over a wide range of pressures and temperatures.

The phase behaviour of the ternary system is derived from the three binary systems. The upper critical pressure of the system carbon dioxide + hexadecane is $p_c(393.2\ K) = 25.6$ MPa and the upper critical pressure of the system carbon dioxide + 1-dodecanol is $p_c(393.2\ K) = 27.5$ MPa. Hexadecane + 1-dodecanol are miscible in all proportions. The ternary critical line connects the two binary critical points. In figure 3 the isothermal system is presented.

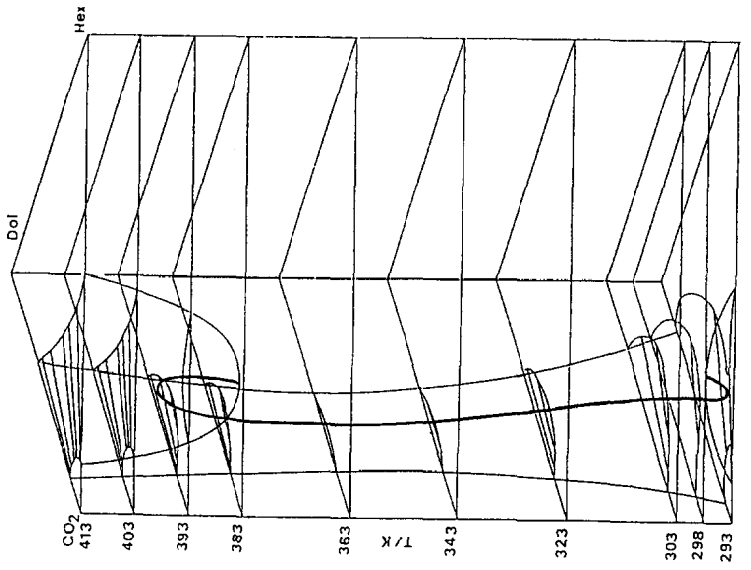



Fig. 4 T(w)-phase prism of the ternary system $CO_2$ + hexadecane + 1-dodecanol at 25 MPa

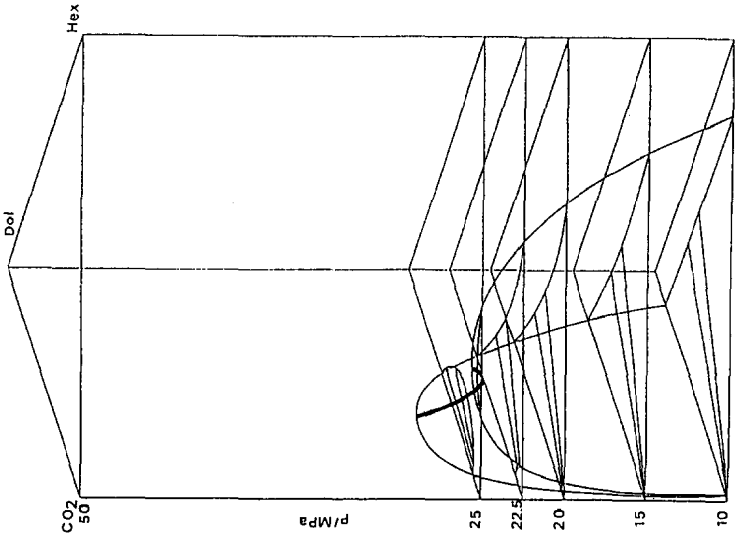



Fig. 3 p(w)-phase prism of the ternary system $CO_2$ + hexadecane + 1-dodecanol at 393.2 K

Very interesting is a pressure minimum on the critical curve. The minimum is found at $p_{c,min}$(393.2 K) = (24.5 ± 0.2) MPa. This system thus represents an example for the so-called "cosolvency" effect.

This effect describes an increased solubility of a mixture of similar substances in a compressed gas like carbon dioxide in comparison with the solubility of each pure component in the same compressed gas. It is evidently caused by the increased entropy of mixing when adding a third substance.

The isobaric binary system carbon dioxide + hexadecane exhibits an upper and a lower critical temperature ($T_u^c$(25 MPa) = (384± 2) K and $T_l^c$(25 MPa) = (298± 2) K). The isobaric binary system carbon dioxide + 1-dodecanol shows a two-phase region of the shape of a band (fig. 2b). The ternary system as a combination of the two binary systems is interesting of the critical curve. The ternary critical curve connects the upper and the lower critical temperature of the binary system carbon dioxide + hexadecane. The cosolvency effect gives origin to a maximum ($T_{c,max}$(25 MPa) ≈ (397± 2) K) and a minimum ($T_{c,min}$(25 MPa) ≈ (293± 2) K) on the critical curve and the homogeneous region resembles a hole in the two-phase body (fig. 4).

carbon dioxide + hexadecane + 1,8-octanediol

The system carbon dioxide + hexadecane + 1,8-octanediol differs from the other ternary systems by the appearance of a three-phase region (fig. 5). For 393.2 K the three- phase region exists below p = 25.1 MPa. The octanediol rich phase has the higher density, the carbon dioxide rich phase is less dense.

The three-phase region is located between three two-phase regions. At p=10 MPa the three-phase region covers 60% of the phase triangle. Each side of the three-phase triangle has been determined separately in the experiments.

carbon dioxide + 1-dodecanol + 1,8-octanediol

Here the ternary critical curve starts at the upper critical pressure of the system carbon dioxide + 1-dodecanol ($p_c$(393.2 K) = 27.5 MPa) and ascends to higher pressures. A heterogeneous area exists all over the investigated pressure interval between 10 and 100 MPa (fig. 6).

Cosolvency is also found here: The ternary critical curve starts with a small slope. Only at high diol-concentrations the critical curve runs steeply to higher pressures.

hexadecane + 1-dodecanol + 1,8-octanediol

In combination of the three binary systems the two-phase region of the ternary system hexadecane + 1-dodecanol + 1,8-octanediol has the shape of a bisected cylinder. There is no detectable dependence on pressure between 10 and 100 MPa (fig. 7).

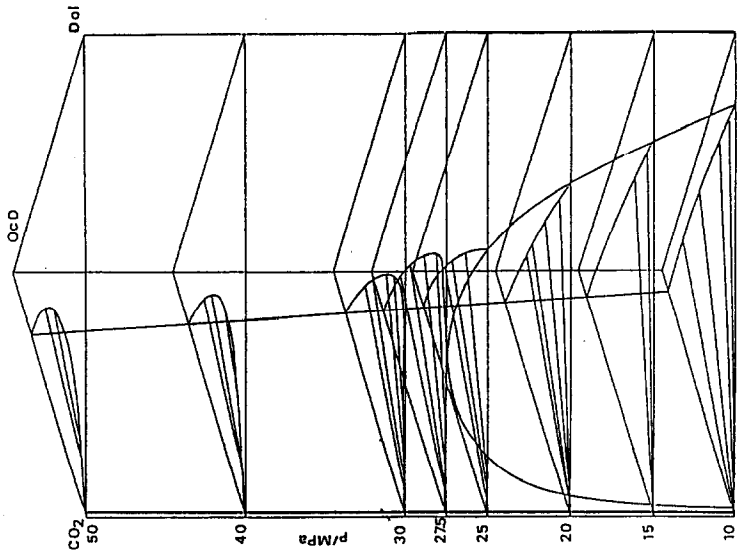



Fig. 6 p(w)-phase prism of the ternary system $CO_2$ + 1-dodecanol + 1,8-octanediol at 393.2 K

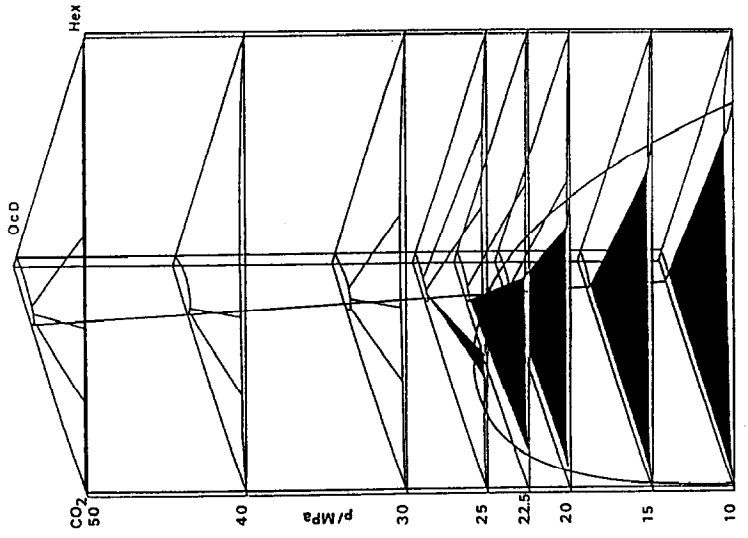



Fig. 5 p(w)-phase prism of the ternary system $CO_2$ + hexadecane + 1,8-octanediol at 393.2 K

carbon dioxide + 1-dodecanol + trifluoromethane

This system was investigated by Katzenski-Ohling (1986, 1987). A new analysis of the data shows that the "island" found in the ternary $p(w_1,w_2)$ section derives only from three equilibrium data of rather low reliability. Without these probably unreliable data the ternary critical curve is not S-shaped. The shape of the two-phase region at lower pressures is not changed.

## CONCLUSION

These systems were primarily investigated to get a quantitative impression of the phase behavior in nearly all combinations of carbon dioxide and the used alkanes and alcohols. Remarkable is the increase of solubility in the ternary systems called cosolvency.

Additionally these systems are within the scope of two quaternary systems of $CO_2$, hexadecane, 1-dodecanol, 1,8-octanediol and dotriacontane.

In a following paper the quaternary systems will be discussed in solubility and separation. Dotriacontane and 1,8-octanediol takes part as useful moderators. For more details and additional results see Spee (1990).

## ACKNOWLEDGEMENTS

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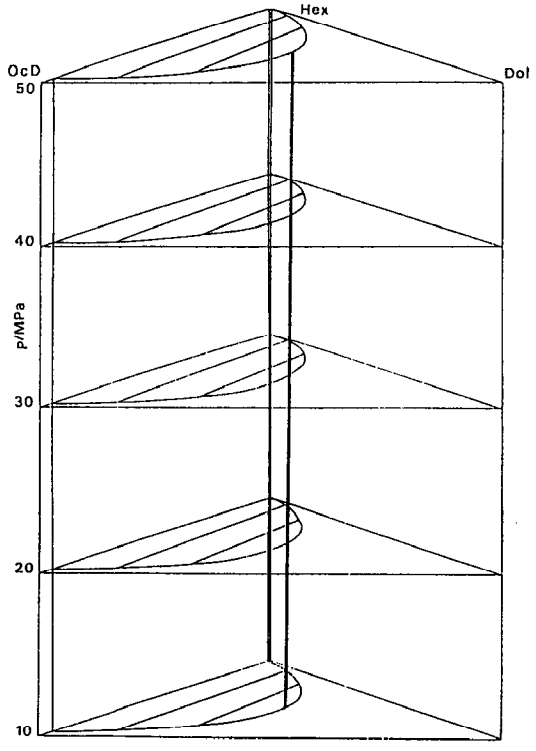



Fig. 7 p(w)-phase prism of the ternary system hexadecane + 1-dodecanol + 1,8-octanediol at 393.2 K

TABLES

$CO_2$(1) + hexadecane(2) (393.2 K)

| p(MPa) | $w_1^g$ | $w_1^l$ | $w_2^g$ | $w_2^l$ |
|---|---|---|---|---|
| 10.1 | 0.991 | 0.161 | 0.009 | 0.839 |
| 13.1 | 0.987 | 0.219 | 0.013 | 0.781 |
| 15.1 | 0.985 | 0.253 | 0.015 | 0.747 |
| 17.4 | 0.968 | 0.307 | 0.032 | 0.693 |
| 20.0 | 0.955 | 0.365 | 0.045 | 0.635 |
| 21.5 | 0.938 | 0.421 | 0.062 | 0.579 |
| 22.5 | 0.932 | 0.438 | 0.068 | 0.562 |
| 23.8 | 0.898 | 0.500 | 0.102 | 0.500 |
| 24.1 | 0.878 | 0.518 | 0.122 | 0.482 |
| 24.0 | 0.883 | 0.507 | 0.117 | 0.493 |
| 25.1 | 0.835 | 0.577 | 0.165 | 0.423 |
| 25.0 | 0.826 | 0.573 | 0.174 | 0.427 |
| 25.0 | 0.851 | 0.577 | 0.149 | 0.423 |
| 25.6 | 0.722 | 0.712 | 0.278 | 0.288 |
| 25.6 | 0.691 | 0.682 | 0.309 | 0.318 |

$CO_2$(1) + hexadecane(2) (25.0 MPa)

| T(K) | $w_1^g$ | $w_1^l$ | $w_2^g$ | $w_2^l$ |
|---|---|---|---|---|
| 294.4 | 0.466 | 0.839 | 0.534 | 0.161 |
| 386.9 | 0.793 | 0.627 | 0.207 | 0.373 |
| 393.3 | 0.851 | 0.577 | 0.149 | 0.423 |
| 403.3 | 0.885 | 0.535 | 0.115 | 0.465 |
| 413.3 | 0.889 | 0.490 | 0.111 | 0.510 |

$CO_2$(1) + 1-dodecanol(3) (393.2 K)

| p(MPa) | $w_1^g$ | $w_1^l$ | $w_3^g$ | $w_3^l$ |
|---|---|---|---|---|
| 10.0 | 0.980 | 0.147 | 0.020 | 0.853 |
| 12.9 | 0.971 | 0.199 | 0.029 | 0.801 |
| 15.0 | 0.974 | 0.227 | 0.026 | 0.773 |
| 17.3 | 0.965 | 0.268 | 0.035 | 0.732 |
| 20.0 | 0.941 | 0.319 | 0.059 | 0.681 |
| 22.5 | 0.936 | 0.386 | 0.064 | 0.614 |
| 25.0 | 0.883 | 0.461 | 0.117 | 0.539 |
| 27.0 | 0.817 | 0.553 | 0.183 | 0.447 |
| 27.5 | 0.647 | 0.652 | 0.353 | 0.348 |

$CO_2$(1) + 1-dodecanol(3) (25.0 MPa)

| T(K) | $w_1^g$ | $w_1^l$ | $w_3^g$ | $w_3^l$ |
|---|---|---|---|---|
| 293.2 | 0.279 | 0.950 | 0.721 | 0.050 |
| 298.2 | 0.296 | 0.926 | 0.704 | 0.074 |
| 303.2 | 0.318 | 0.907 | 0.682 | 0.093 |
| 313.2 | 0.361 | 0.907 | 0.639 | 0.093 |
| 323.1 | 0.846 | 0.402 | 0.154 | 0.598 |
| 333.2 | 0.805 | 0.439 | 0.195 | 0.561 |
| 343.0 | 0.788 | 0.466 | 0.212 | 0.534 |
| 352.6 | 0.792 | 0.482 | 0.208 | 0.518 |
| 363.1 | 0.811 | 0.501 | 0.189 | 0.499 |
| 373.2 | 0.812 | 0.473 | 0.188 | 0.527 |
| 383.2 | 0.826 | 0.469 | 0.174 | 0.531 |
| 393.2 | 0.883 | 0.461 | 0.117 | 0.539 |

$CO_2$(1) + dotriacontane(5) (393.2 K)

| p(MPa) | $w_1^g$ | $w_1^l$ | $w_5^g$ | $w_5^l$ |
|---|---|---|---|---|
| 10.0 | 0.997 | 0.170 | 0.003 | 0.830 |
| 20.0 | 0.996 | 0.240 | 0.004 | 0.760 |
| 20.0 | 0.997 | 0.243 | 0.003 | 0.757 |
| 25.0 | 0.992 | 0.262 | 0.008 | 0.738 |
| 30.0 | 0.982 | 0.291 | 0.018 | 0.709 |
| 40.0 | 0.972 | 0.389 | 0.028 | 0.611 |
| 50.0 | 0.936 | 0.491 | 0.064 | 0.509 |
| 60.0 | 0.864 | 0.554 | 0.136 | 0.446 |
| 63.0 | 0.616 | 0.827 | 0.384 | 0.173 |
| 63.0 | 0.629 | 0.830 | 0.371 | 0.170 |

$CO_2$(1) + 1,8-octanediol(4) (393.2 K)

| p(MPa) | $w_1^g$ | $w_1^l$ | $w_4^g$ | $w_4^l$ |
|---|---|---|---|---|
| 10.0 | 0.994 | 0.086 | 0.006 | 0.914 |
| 20.2 | 0.997 | 0.136 | 0.003 | 0.864 |
| 22.5 | 0.994 | 0.145 | 0.006 | 0.855 |
| 25.0 | 0.989 | 0.153 | 0.011 | 0.847 |
| 27.5 | 0.987 | 0.163 | 0.013 | 0.837 |
| 30.2 | 0.986 | 0.170 | 0.014 | 0.830 |
| 40.5 | 0.982 | 0.213 | 0.018 | 0.787 |
| 50.0 | 0.972 | 0.240 | 0.028 | 0.760 |
| 51.0 | 0.964 | 0.230 | 0.036 | 0.770 |
| 60.0 | 0.955 | 0.262 | 0.045 | 0.738 |
| 65.0 | 0.950 | 0.268 | 0.050 | 0.732 |
| 70.4 | 0.947 | 0.275 | 0.053 | 0.725 |
| 80.0 | 0.937 | 0.289 | 0.063 | 0.711 |
| 87.0 | 0.931 | 0.301 | 0.069 | 0.699 |
| 95.0 | 0.915 | 0.307 | 0.085 | 0.693 |
| 98.0 | 0.912 | 0.307 | 0.088 | 0.693 |

hexadecane(2) + 1,8-octanediol(4) (393.2 K)

| p(MPa) | $w_2^g$ | $w_2^l$ | $w_4^g$ | $w_4^l$ |
|---|---|---|---|---|
| 10.0 | 0.991 | 0.043 | 0.009 | 0.957 |
| 20.0 | 0.990 | 0.042 | 0.010 | 0.958 |
| 30.0 | 0.991 | 0.040 | 0.009 | 0.960 |
| 40.0 | 0.993 | 0.043 | 0.007 | 0.957 |
| 50.0 | 0.992 | 0.043 | 0.008 | 0.957 |
| 70.0 | 0.991 | 0.043 | 0.009 | 0.957 |

$CO_2$(1) + hexadecane(2) + 1-dodecanol(3) (393.2 K)

| p(MPa) | $w_1^g$ | $w_1^l$ | $w_2^g$ | $w_2^l$ | $w_3^g$ | $w_3^l$ |
|---|---|---|---|---|---|---|
| 10.0 | 0.989 | 0.171 | 0.004 | 0.278 | 0.007 | 0.551 |
| 10.0 | 0.993 | 0.168 | 0.004 | 0.416 | 0.003 | 0.416 |
| 15.0 | 0.982 | 0.267 | 0.007 | 0.245 | 0.011 | 0.488 |
| 15.0 | 0.975 | 0.267 | 0.013 | 0.366 | 0.012 | 0.367 |
| 20.0 | 0.940 | 0.390 | 0.021 | 0.204 | 0.039 | 0.406 |
| 20.0 | 0.948 | 0.398 | 0.026 | 0.303 | 0.026 | 0.299 |
| 22.5 | 0.908 | 0.474 | 0.033 | 0.176 | 0.059 | 0.350 |
| 22.5 | 0.913 | 0.478 | 0.032 | 0.176 | 0.055 | 0.346 |
| 22.5 | 0.899 | 0.490 | 0.049 | 0.256 | 0.052 | 0.254 |
| 24.0 | 0.828 | 0.544 | 0.061 | 0.153 | 0.111 | 0.303 |
| 24.0 | 0.831 | 0.569 | 0.085 | 0.217 | 0.084 | 0.214 |
| 24.6 | 0.691 | 0.689 | 0.155 | 0.156 | 0.154 | 0.155 |
| 25.1 | 0.832 | 0.475 | 0.015 | 0.040 | 0.153 | 0.485 |
| 25.0 | 0.822 | 0.522 | 0.037 | 0.103 | 0.141 | 0.375 |
| 25.0 | 0.657 | 0.659 | 0.116 | 0.115 | 0.227 | 0.226 |

hexadecane(2) + 1-dodecanol(3) + 1,8-octanediol(4) (393.2 K)

| p(MPa) | $w_2^g$ | $w_2^l$ | $w_3^g$ | $w_3^l$ | $w_4^g$ | $w_4^l$ |
|---|---|---|---|---|---|---|
| 10.0 | 0.886 | 0.074 | 0.090 | 0.133 | 0.024 | 0.793 |
| 10.0 | 0.695 | 0.166 | 0.215 | 0.271 | 0.090 | 0.563 |
| 20.0 | 0.890 | 0.074 | 0.088 | 0.136 | 0.022 | 0.790 |
| 20.0 | 0.704 | 0.159 | 0.212 | 0.263 | 0.084 | 0.578 |
| 30.0 | 0.894 | 0.074 | 0.086 | 0.137 | 0.020 | 0.789 |
| 30.0 | 0.735 | 0.170 | 0.210 | 0.275 | 0.055 | 0.555 |
| 40.0 | 0.897 | 0.077 | 0.084 | 0.137 | 0.019 | 0.786 |
| 40.0 | 0.740 | 0.160 | 0.207 | 0.275 | 0.053 | 0.565 |
| 50.0 | 0.902 | 0.076 | 0.083 | 0.138 | 0.015 | 0.786 |
| 50.0 | 0.752 | 0.164 | 0.202 | 0.268 | 0.046 | 0.568 |
| 65.0 | 0.905 | 0.079 | 0.080 | 0.139 | 0.015 | 0.782 |
| 65.0 | 0.752 | 0.150 | 0.196 | 0.256 | 0.052 | 0.594 |
| 80.0 | 0.909 | 0.079 | 0.078 | 0.139 | 0.013 | 0.782 |
| 80.0 | 0.762 | 0.151 | 0.192 | 0.265 | 0.046 | 0.584 |
| 95.0 | 0.906 | 0.081 | 0.077 | 0.137 | 0.017 | 0.782 |

$CO_2(1)$ + hexadecane(2) + 1-dodecanol(3) (25.0 MPa)

| T(K) | $w_1^g$ | $w_1^l$ | $w_2^g$ | $w_2^l$ | $w_3^g$ | $w_3^l$ |
|---|---|---|---|---|---|---|
| 293.5 | 0.418 | 0.837 | 0.308 | 0.094 | 0.274 | 0.069 |
| 293.2 | 0.693 | 0.816 | 0.254 | 0.155 | 0.053 | 0.029 |
| 298.9 | 0.406 | 0.833 | 0.224 | 0.069 | 0.370 | 0.098 |
| 298.5 | 0.482 | 0.779 | 0.279 | 0.122 | 0.239 | 0.099 |
| 303.5 | 0.354 | 0.837 | 0.115 | 0.042 | 0.531 | 0.121 |
| 303.6 | 0.454 | 0.786 | 0.209 | 0.091 | 0.337 | 0.123 |
| 313.3 | 0.371 | 0.824 | 0.040 | 0.016 | 0.589 | 0.160 |
| 313.3 | 0.410 | 0.806 | 0.105 | 0.045 | 0.485 | 0.149 |
| 323.3 | 0.832 | 0.422 | 0.018 | 0.037 | 0.150 | 0.541 |
| 323.3 | 0.767 | 0.523 | 0.052 | 0.088 | 0.181 | 0.389 |
| 333.3 | 0.809 | 0.481 | 0.018 | 0.035 | 0.173 | 0.484 |
| 343.3 | 0.699 | 0.532 | 0.025 | 0.031 | 0.276 | 0.437 |
| 353.2 | 0.638 | 0.643 | 0.025 | 0.026 | 0.337 | 0.331 |
| 363.2 | 0.646 | 0.580 | 0.031 | 0.033 | 0.323 | 0.387 |
| 373.1 | 0.759 | 0.538 | 0.018 | 0.033 | 0.223 | 0.429 |
| 383.3 | 0.818 | 0.488 | 0.017 | 0.039 | 0.165 | 0.473 |
| 383.1 | 0.734 | 0.591 | 0.056 | 0.084 | 0.210 | 0.325 |
| 393.2 | 0.832 | 0.475 | 0.015 | 0.040 | 0.153 | 0.485 |
| 393.3 | 0.822 | 0.522 | 0.037 | 0.103 | 0.141 | 0.375 |
| 398. | 0.797 | 0.598 | 0.083 | 0.178 | 0.120 | 0.224 |
| 397.7 | 0.804 | 0.621 | 0.162 | 0.312 | 0.034 | 0.067 |
| 403.4 | 0.848 | 0.439 | 0.014 | 0.041 | 0.138 | 0.520 |
| 403.4 | 0.834 | 0.484 | 0.034 | 0.109 | 0.132 | 0.407 |
| 403.5 | 0.819 | 0.536 | 0.076 | 0.207 | 0.105 | 0.257 |
| 403.3 | 0.815 | 0.574 | 0.100 | 0.250 | 0.085 | 0.176 |
| 403.2 | 0.837 | 0.571 | 0.137 | 0.355 | 0.026 | 0.074 |
| 413.1 | 0.862 | 0.429 | 0.011 | 0.045 | 0.127 | 0.526 |
| 413.1 | 0.845 | 0.437 | 0.031 | 0.121 | 0.124 | 0.442 |
| 413.3 | 0.852 | 0.511 | 0.061 | 0.219 | 0.087 | 0.270 |
| 413.3 | 0.850 | 0.512 | 0.078 | 0.286 | 0.072 | 0.202 |
| 413.5 | 0.864 | 0.490 | 0.118 | 0.422 | 0.018 | 0.088 |

$CO_2(1)$ + hexadecane(2) + 1,8-octanediol(4) (393.2 K)

| p(MPa) | $w_1^g$ | $w_1^l$ | $w_2^g$ | $w_2^l$ | $w_4^g$ | $w_4^l$ |
|---|---|---|---|---|---|---|
| 10.0 | 0.982 | 0.088 | 0.004 | 0.037 | 0.014 | 0.875 |
| 10.0 | 0.159 | 0.087 | 0.728 | 0.040 | 0.113 | 0.873 |
| 10.0 | 0.988 | 0.101 | 0.005 | 0.042 | 0.007 | 0.857 |
| 10.0 | 0.964 | 0.108 | 0.004 | 0.046 | 0.032 | 0.846 |
| 10.0 | 0.998 | 0.166 | 0.002 | 0.723 | 0.000 | 0.111 |
| 15.0 | 0.981 | 0.130 | 0.012 | 0.037 | 0.007 | 0.833 |
| 15.0 | 0.963 | 0.135 | 0.004 | 0.039 | 0.033 | 0.826 |
| 15.0 | 0.284 | 0.138 | 0.676 | 0.045 | 0.040 | 0.817 |
| 15.0 | 0.983 | 0.277 | 0.016 | 0.698 | 0.001 | 0.025 |
| 20.0 | 0.956 | 0.160 | 0.037 | 0.037 | 0.007 | 0.803 |
| 20.0 | 0.947 | 0.171 | 0.038 | 0.042 | 0.015 | 0.787 |
| 20.0 | 0.327 | 0.168 | 0.644 | 0.044 | 0.029 | 0.788 |
| 20.0 | 0.960 | 0.407 | 0.039 | 0.556 | 0.001 | 0.037 |
| 22.5 | 0.920 | 0.159 | 0.077 | 0.036 | 0.003 | 0.805 |
| 22.6 | 0.921 | 0.163 | 0.077 | 0.036 | 0.002 | 0.801 |
| 22.5 | 0.342 | 0.173 | 0.625 | 0.041 | 0.033 | 0.786 |
| 22.5 | 0.513 | 0.190 | 0.447 | 0.042 | 0.040 | 0.768 |
| 22.5 | 0.910 | 0.504 | 0.088 | 0.446 | 0.002 | 0.050 |
| 22.5 | 0.918 | 0.500 | 0.080 | 0.456 | 0.002 | 0.044 |
| 24.0 | 0.859 | 0.177 | 0.129 | 0.038 | 0.012 | 0.785 |
| 24.0 | 0.561 | 0.214 | 0.379 | 0.037 | 0.060 | 0.749 |
| 24.0 | 0.863 | 0.595 | 0.129 | 0.370 | 0.008 | 0.035 |
| 25.0 | 0.735 | 0.172 | 0.245 | 0.036 | 0.020 | 0.792 |
| 25.0 | 0.343 | 0.161 | 0.600 | 0.044 | 0.057 | 0.795 |
| 25.0 | 0.548 | 0.200 | 0.379 | 0.049 | 0.073 | 0.751 |
| 25.0 | 0.691 | 0.685 | 0.284 | 0.289 | 0.025 | 0.026 |
| 27.5 | 0.736 | 0.185 | 0.235 | 0.033 | 0.029 | 0.782 |

$CO_2(1)$ + hexadecane(2) + 1,8-octanediol(4) (393.2 K)

| p(MPa) | $w_1^g$ | $w_1^l$ | $w_2^g$ | $w_2^l$ | $w_4^g$ | $w_4^l$ |
|---|---|---|---|---|---|---|
| 27.5 | 0.579 | 0.226 | 0.397 | 0.041 | 0.024 | 0.733 |
| 30.0 | 0.728 | 0.213 | 0.229 | 0.035 | 0.043 | 0.752 |
| 30.0 | 0.556 | 0.190 | 0.367 | 0.041 | 0.077 | 0.769 |
| 30.0 | 0.363 | 0.160 | 0.602 | 0.049 | 0.035 | 0.791 |
| 40.0 | 0.734 | 0.217 | 0.204 | 0.031 | 0.062 | 0.752 |
| 40.0 | 0.570 | 0.215 | 0.354 | 0.039 | 0.076 | 0.746 |
| 50.0 | 0.758 | 0.256 | 0.193 | 0.030 | 0.049 | 0.714 |
| 50.0 | 0.602 | 0.241 | 0.351 | 0.043 | 0.047 | 0.716 |
| 50.0 | 0.419 | 0.216 | 0.549 | 0.048 | 0.032 | 0.736 |
| 65.0 | 0.614 | 0.249 | 0.317 | 0.043 | 0.069 | 0.708 |
| 65.0 | 0.439 | 0.209 | 0.523 | 0.060 | 0.038 | 0.731 |
| 65.0 | 0.770 | 0.765 | 0.179 | 0.182 | 0.051 | 0.053 |
| 80.0 | 0.629 | 0.291 | 0.298 | 0.048 | 0.073 | 0.661 |
| 80.0 | 0.441 | 0.245 | 0.521 | 0.059 | 0.038 | 0.696 |
| 95.0 | 0.454 | 0.239 | 0.505 | 0.052 | 0.041 | 0.709 |
| 95.0 | 0.635 | 0.301 | 0.302 | 0.050 | 0.063 | 0.649 |

$CO_2(1)$ + 1-dodecanol(3) + 1,8-octanediol(4) (393.2 K)

| p(MPa) | $w_1^g$ | $w_1^l$ | $w_3^g$ | $w_3^l$ | $w_4^g$ | $w_4^l$ |
|---|---|---|---|---|---|---|
| 10.0 | 0.963 | 0.127 | 0.028 | 0.671 | 0.009 | 0.202 |
| 10.2 | 0.984 | 0.092 | 0.007 | 0.228 | 0.009 | 0.680 |
| 10.2 | 0.972 | 0.114 | 0.021 | 0.508 | 0.007 | 0.378 |
| 10.0 | 0.990 | 0.145 | 0.010 | 0.779 | 0.000 | 0.076 |
| 15.1 | 0.976 | 0.204 | 0.024 | 0.614 | 0.000 | 0.182 |
| 15.2 | 0.975 | 0.233 | 0.025 | 0.702 | 0.000 | 0.065 |
| 20.3 | 0.958 | 0.176 | 0.024 | 0.237 | 0.018 | 0.587 |
| 20.1 | 0.970 | 0.194 | 0.020 | 0.305 | 0.010 | 0.501 |
| 20.4 | 0.957 | 0.247 | 0.035 | 0.430 | 0.008 | 0.323 |
| 20.2 | 0.966 | 0.292 | 0.029 | 0.551 | 0.005 | 0.157 |
| 20.1 | 0.962 | 0.313 | 0.031 | 0.627 | 0.007 | 0.060 |
| 22.5 | 0.969 | 0.188 | 0.019 | 0.232 | 0.012 | 0.580 |
| 22.5 | 0.964 | 0.215 | 0.025 | 0.293 | 0.011 | 0.492 |
| 22.5 | 0.949 | 0.274 | 0.042 | 0.414 | 0.009 | 0.312 |
| 22.5 | 0.951 | 0.343 | 0.044 | 0.511 | 0.005 | 0.146 |
| 22.5 | 0.946 | 0.368 | 0.049 | 0.582 | 0.005 | 0.050 |
| 25.0 | 0.965 | 0.210 | 0.028 | 0.224 | 0.007 | 0.566 |
| 25.2 | 0.955 | 0.240 | 0.033 | 0.280 | 0.012 | 0.480 |
| 25.3 | 0.938 | 0.300 | 0.052 | 0.399 | 0.010 | 0.301 |
| 25.5 | 0.925 | 0.395 | 0.068 | 0.471 | 0.007 | 0.134 |
| 25.4 | 0.907 | 0.442 | 0.091 | 0.518 | 0.002 | 0.040 |
| 27.6 | 0.918 | 0.243 | 0.055 | 0.273 | 0.027 | 0.484 |
| 27.5 | 0.835 | 0.390 | 0.143 | 0.475 | 0.022 | 0.135 |
| 27.8 | 0.942 | 0.213 | 0.038 | 0.217 | 0.020 | 0.570 |
| 27.7 | 0.899 | 0.320 | 0.081 | 0.383 | 0.020 | 0.297 |
| 27.5 | 0.834 | 0.441 | 0.157 | 0.499 | 0.009 | 0.060 |
| 27.5 | 0.811 | 0.477 | 0.172 | 0.489 | 0.017 | 0.034 |
| 28.6 | 0.639 | 0.635 | 0.334 | 0.337 | 0.027 | 0.028 |
| 30.0 | 0.914 | 0.256 | 0.051 | 0.202 | 0.035 | 0.542 |
| 30.1 | 0.912 | 0.268 | 0.061 | 0.260 | 0.027 | 0.472 |
| 30.2 | 0.847 | 0.377 | 0.113 | 0.350 | 0.040 | 0.273 |
| 30.0 | 0.839 | 0.444 | 0.143 | 0.432 | 0.018 | 0.124 |
| 32.4 | 0.637 | 0.637 | 0.286 | 0.286 | 0.077 | 0.077 |
| 40.3 | 0.874 | 0.286 | 0.083 | 0.170 | 0.043 | 0.544 |
| 40.2 | 0.866 | 0.310 | 0.091 | 0.225 | 0.043 | 0.465 |
| 40.2 | 0.629 | 0.458 | 0.229 | 0.290 | 0.142 | 0.252 |
| 43.8 | 0.630 | 0.642 | 0.220 | 0.209 | 0.150 | 0.149 |
| 50.3 | 0.839 | 0.301 | 0.091 | 0.155 | 0.070 | 0.544 |
| 50.3 | 0.814 | 0.347 | 0.109 | 0.201 | 0.077 | 0.452 |
| 65.4 | 0.831 | 0.351 | 0.085 | 0.133 | 0.084 | 0.516 |
| 65.1 | 0.747 | 0.407 | 0.123 | 0.172 | 0.130 | 0.421 |
| 77.8 | 0.674 | 0.667 | 0.129 | 0.126 | 0.197 | 0.207 |
| 80.2 | 0.787 | 0.398 | 0.091 | 0.121 | 0.122 | 0.481 |
| 95.0 | 0.751 | 0.454 | 0.083 | 0.119 | 0.166 | 0.427 |

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